

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 50%

QUESTIONS: 5 Total Marks:50 Annexure A

Data Interpretation

Code of Conduct:

I VALLAPU MOHAN KUMAR (192524225) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Mohan Kumar

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	

Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	
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Annexure A

Data Interpretation

Question 1: Hough Transform & Shape Detection

Scenario: Detect linear features (roads, rivers) in satellite images. **Input Data:**

- High-resolution grayscale satellite images of urban and rural areas (e.g., 1024×1024 pixels, TIFF or PNG)
- Images contain noise (clouds, shadows) and partial occlusions
- Multiple linear features (roads, rivers, canals) with intersections
- Optional synthetic dataset: images with drawn lines at random angles and lengths to test Hough Transform

Question 2: PCA & Shape Priors for Object Recognition

Scenario: Segment organs in MRI scans using shape priors. **Input Data:**

- MRI slices (grayscale, 512×512 pixels) of organs like liver or heart
- Training set: 50–100 annotated organ masks for PCA-based shape model construction
- Test set: 10–20 MRI images with partial organ boundaries or noise
- File formats: DICOM or PNG

Question 3: Machine Learning Methods (SVM, HMM, GMM, EM)

Scenario: Human activity recognition from video feature trajectories. **Input Data:**

- Video sequences (30–60 FPS) of activities like walking, running, jumping (e.g., 640×480 pixels, MP4)
- Feature trajectories extracted per frame: joint positions (x, y coordinates) or optical flow vectors
- Training labels: activity per video (e.g., walking, jumping)
- Optional synthetic data: CSV file with simulated joint coordinates over time

Question 4: Deep Learning (CNNs, Transformers, GANs)

Scenario: Detect rare obstacles in urban self-driving car images. **Input Data:**

- A. RGB camera images of urban streets (1024×768 pixels, JPG/PNG)
- B. Labeled dataset with bounding boxes for obstacles (rare objects, e.g., traffic cones, animals)
- C. Small dataset: 500–1000 labeled images for real obstacles
- D.

Optional synthetic images generated by GAN for data augmentation

Question 5: Graph-Based Learning for Shape Correspondence

Scenario: Find correspondences between 3D human face meshes. **Input Data:**

- A. 3D meshes of human faces (OBJ or PLY format) with varying vertex counts (5,000–50,000 vertices)
- B. Training set: 20–50 face meshes with annotated landmark correspondences
- C. Test set: 5–10 unseen face meshes
- D. Optional: Graph adjacency matrices representing each mesh for input to graph-based algorithms